

Task 2: OS Security Fundamentals (Linux & Windows)

- **FULL WINDOWS WORK**

- user accounts & privileges
- permissions/access control
- Windows Defender
- Firewall rules
- running processes/services
- disabling unnecessary services
- OS hardening checklist

Tools Used (Windows)

- Windows Security (Microsoft Defender)
- Windows Defender Firewall
- Task Manager
- Services (services.msc)
- Local Users and Groups (lusrmgr.msc)
- Command Prompt / PowerShell

Check Windows Version + System Info



Also open CMD:

Systeminfo

```
C:\Users\hari6>systeminfo

Host Name:                HARIAHARAN4C
OS Name:                  Microsoft Windows [1 Home Single Language]
OS Version:               10.0.22000 N/A Build 22000
OS Manufacturer:         Microsoft Corporation
OS Configuration:         Standard installation
OS Build Type:             Multiprocessor Free
Registered Owner:         hari6@h21poutlook.com
Registered Organization:   H21P@outlook.com
Product ID:                88556-2200
Original Install Date:     26-06-2025, 22:09:32
System Boot Time:          12-02-2025, 18:06:42
System Manufacturer:       Acer
System Model:              Extensa 215-56
System Type:               x64-based PC
Processor(s):              1 Processor(s) Installed.
                          [01]: Intel(R) Family 8 Model 148 Stepping 1 GenuineIntel ~2995 Mhz
BIOS Version:              Insyde Corp. V1.26, 14-03-2022
Windows Directory:         C:\Windows
System Directory:          C:\Windows\System32
Boot Device:               \Device\HarddiskVolume1
System Locale:              en-IN;English (United Kingdom)
Input Locale:              00000905
Time Zone:                 (UTC+05:30) Chennai, Kolkata, Mumbai, New Delhi
Total Physical Memory:     7.975 MB
Available Physical Memory: 705 MB
Virtual Memory: Max Size:  15,526 MB
Virtual Memory: Available: 973 MB
Virtual Memory: In Use:    14,553 MB
Page File Location(s):     C:\pagefile.sys
Domain:                    \HARIAHARAN4C
Logon Server:              h21p@outlook.com
Netlogon Server:           h21p@outlook.com

Network Card(s):           4 NIC(s) Installed.
                          [01]: VirtualBox Host-Only Ethernet Adapter
                              Connection Name: Ethernet 3
                              DHCP Enabled:    No
                              IP address(es):  [01]: 192.168.44.1
                              [02]: ::::: IPv6 ::
                          [02]: Bluetooth Usb x (Personal Area Network)
                              Connection Name: Bluetooth Network Connection
                              Status:          Media disconnected
                          [03]: Intel(R) Wireless-AC 9462
                              Connection Name: WiFi
                              DHCP Enabled:     Yes
                              DHCP Server:     192.168.43.1
                              IP address(es):  [01]: 192.168.43.121
                              [02]: f4b8:a622:a652:a657:a899
                              [03]: 2481:4000:4499:a65c:508d:8ac9:508c:c07a
                              [04]: 2481:4000:4499:a65c:508d:18ca:9047:f977
                          [04]: Realtek PCIe GbE Family Controller
                              Connection Name: Ethernet
                              Status:          Media disconnected

Virtualization-based security: Status: Pending
                               Required Security Properties:
                               Available Security Properties:
                                   Basic Virtualization Support
                                   DMA Protection
                                   SME Security Mitigations 1.0
                                   Mde Based Execution Control
```

User Accounts & Privileges

```
C:\Users\hari6>whoami
hariaharan4c\hari6
```

```
C:\Users\hari6>whoami /groups

GROUP INFORMATION
-----
Group Name                                     Type                SID                                     Attributes
-----
Everyone                                     Label               S-1-16-0180
Administrators                               Well-known group    S-1-1-0
NT AUTHORITY\LOCAL account and member of Administrators group Well-known group S-1-5-114
BUILTIN\Administrators                     Alias               S-1-5-32-544
BUILTIN\Users                             Alias               S-1-5-32-545
NT AUTHORITY\INTERACTIVE                   Well-known group    S-1-5-4
LOCAL Users                               Well-known group    S-1-5-21
NT AUTHORITY\Authenticated Users           Well-known group    S-1-5-31
NT AUTHORITY\This Organization              Well-known group    S-1-5-32
NT AUTHORITY\Account\HARI6@h21poutlook.com User                S-1-11-94-3623034031-32344-18804-28A17222817-13971811943-28131870614-4286818067-3787385233-2788437382-3781395532
NT AUTHORITY\LOCAL account                 Well-known group    S-1-5-113
LOCAL                                       Well-known group    S-1-5-2-4
NT AUTHORITY\Cloud Account Authentication    Well-known group    S-1-5-66-30
NT AUTHORITY\Cloud Account Authentication    Well-known group    S-1-5-66-30
C:\Users\hari6\
```

```
C:\Users\hari6>whoami /priv
```

```
PRIVILEGES INFORMATION
-----
Privilege Name      Description                State
-----
SeLockMemoryPrivilege Lock pages in memory      Disabled
SeShutdownPrivilege Shut down the system      Disabled
SeChangeNotifyPrivilege Bypass traverse checking   Enabled
SeUndockPrivilege Remove computer from docking station Disabled
SeIncreaseWorkingSetPrivilege Increase a process working set Disabled
SeTimeZonePrivilege Change the time zone       Disabled
```

```
User accounts for \\HARIAHARAN4C
```

```
Administrator      DefaultAccount
hari6                WDAGUtilityAccount
The command completed successfully.
```

Check admin users:

```
C:\Users\hari6>net localgroup administrators
Alias name     administrators
Comment       Administrators have complete and unrestricted access to the computer/domain

Members

-----
Administrator
hari6
The command completed successfully.
```

Access Control & File Permissions

Command: check NTFS permissions

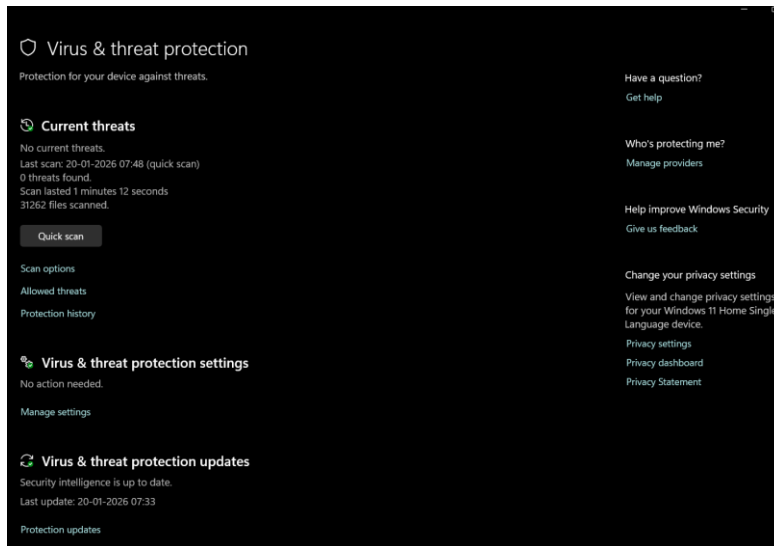
```
C:\temp>icacls "C:\temp"
C:\temp BUILTIN\Administrators:(I)(OI)(CI)(F)
NT AUTHORITY\SYSTEM:(I)(OI)(CI)(F)
BUILTIN\Users:(I)(OI)(CI)(RX)
NT AUTHORITY\Authenticated Users:(I)(M)
NT AUTHORITY\Authenticated Users:(I)(OI)(CI)(IO)(M)

Successfully processed 1 files; Failed processing 0 files
```

Windows Defender (Antivirus Protection)

Go to:

Windows Security → Virus & threat protection



PowerShell commands

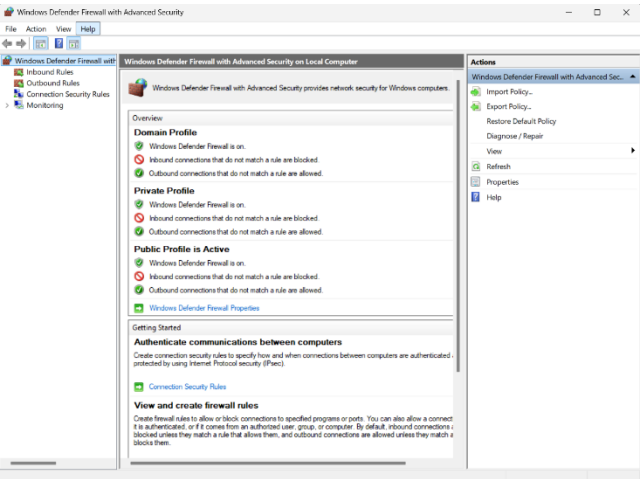
```
$ C:\Users\hari6> Get-MpComputerStatus

EngineVersion           : 1.1.25110.1
MpProductVersion        : 4.18.25110.6
MpRunningMode           : Normal
MpServiceEnabled        : True
MpServiceVersion        : 4.18.25110.6
MpSignatureEnabled      : True
MpSignatureAge           : 15
MpSignatureLastUpdated   : 06-01-2026 06:58:45
MpSignatureVersion      : 1.443.506.0
MpSignatureAge           : 15
MpSignatureLastUpdated   : 06-01-2026 06:58:45
MpSignatureVersion      : 1.443.506.0
BehaviorMonitorEnabled   : True
ComputerID              : 3CCFE33F-BB66-4CA6-B149-14FA818439B5
ComputerState           : 0
DefenderSignaturesOutOfDate : True
DeviceControlDefaultEnforcement : 01-01-1601 05:30:00
DeviceControlPoliciesLastUpdated : Disabled
DeviceControlState      : Disabled
FullScanAge             : 4294967295
FullScanEndTime         : 
FullScanOverdue         : False
FullScanRequired        : False
FullScanSignatureVersion : 
FullScanStartTime       : 
InitializationProgress   : ServiceStartedSuccessfully
IoavProtectionEnabled    : True
IsTamperProtected        : True
IsVirtualMachine         : False
LastFullScanSource       : 0
LastQuickScanSource      : 2
IISEnabled              : True
IISEngineVersion         : 1.1.25110.1
IISSignatureAge          : 15
IISSignatureLastUpdated   : 06-01-2026 06:58:45
IISSignatureVersion      : 1.443.506.0
MpAccessProtectionEnabled : True
ProductStatus           : 524512
QuickScanAge            : 15
QuickScanEndTime        : 04-01-2026 22:59:53
QuickScanOverdue        : True
QuickScanSignatureVersion : 1.443.358.0
QuickScanStartTime       : 04-01-2026 22:58:36
RealTimeProtectionEnabled : True
RealTimeScanDirection   : 0
RebootRequired           : False
SmartAppControlExpiration : 
SmartAppControlState     : Off
TamperProtectionSource    : Signatures
DTCapable               : N/A
DTMode                   : N/A
DTSilentType             : N/A
DTStatus                 : N/A
DTTelemetry              : N/A
DoubleShootingDailyMaxQuota : 
DoubleShootingDailyQuotaLeft : 
DoubleShootingEndTime    : 
DoubleShootingExpirationLeft : 
DoubleShootingMode       : 
DoubleShootingModeSource  : 
DoubleShootingQuotaResetTime : 
DoubleShootingStartTime   : 
$ComputerName            :
```

Update signatures:

Update-MpSignature

Windows Defender Firewall



CMD commands

```
C:\temp>netsh advfirewall show allprofiles

Domain Profile Settings:
-----
State: ON
Firewall Policy: BlockInbound,AllowOutbound
LocalFirewallRules: N/A (GPO-store only)
LocalConnectionRules: N/A (GPO-store only)
InboundIsolation: Enable
RemoteManagement: Disable
UnicastResponseToMulticast: Enable

Logging:
LogAllowedConnections: Disable
LogDroppedConnections: Disable
Fileshare: %systemroot%\system32\LogFiles\Firewall\pfirewall.log
MaxFileSize: 4096

Private Profile Settings:
-----
State: ON
Firewall Policy: BlockInbound,AllowOutbound
LocalFirewallRules: N/A (GPO-store only)
LocalConnectionRules: N/A (GPO-store only)
InboundIsolation: Enable
RemoteManagement: Disable
UnicastResponseToMulticast: Enable

Logging:
LogAllowedConnections: Disable
LogDroppedConnections: Disable
Fileshare: %systemroot%\system32\LogFiles\Firewall\pfirewall.log
MaxFileSize: 4096

Public Profile Settings:
-----
State: ON
Firewall Policy: BlockInbound,AllowOutbound
LocalFirewallRules: N/A (GPO-store only)
LocalConnectionRules: N/A (GPO-store only)
InboundIsolation: Enable
RemoteManagement: Disable
UnicastResponseToMulticast: Enable

Logging:
LogAllowedConnections: Disable
LogDroppedConnections: Disable
Fileshare: %systemroot%\system32\LogFiles\Firewall\pfirewall.log
MaxFileSize: 4096

On.
```

Identify Running Processes

Show Process list via

```
C:\temp>tasklist

Image Name PID Session Name Session# Mem Usage
-----
System Idle Process 0 Services 0 0 K
System 4 Services 0 132 K
Secure System 92 Services 0 48,824 K
Registry 136 Services 0 14,536 K
smss.exe 580 Services 0 428 K
csrss.exe 628 Services 0 2,160 K
wininit.exe 960 Services 0 452 K
services.exe 664 Services 0 6,144 K
lsass.exe 540 Services 0 40 K
lsass.exe 692 Services 0 12,632 K
svchost.exe 1280 Services 0 20,496 K
cmd.exe 1212 Services 0 140 K
smss.exe 1296 Services 0 4 K
svchost.exe 1340 Services 0 14,588 K
svchost.exe 1412 Services 0 2,876 K
svchost.exe 1668 Services 0 2,140 K
svchost.exe 1628 Services 0 784 K
svchost.exe 1768 Services 0 4,716 K
svchost.exe 1772 Services 0 1,984 K
svchost.exe 1820 Services 0 2,180 K
svchost.exe 1824 Services 0 7,188 K
svchost.exe 1852 Services 0 5,280 K
svchost.exe 1956 Services 0 6,356 K
svchost.exe 1968 Services 0 2,380 K
svchost.exe 1988 Services 0 2,128 K
svchost.exe 1996 Services 0 2,572 K
svchost.exe 768 Services 0 3,560 K
svchost.exe 488 Services 0 1,968 K
IntelCpMPCSvc.exe 2192 Services 0 160 K
svchost.exe 2208 Services 0 6,464 K
svchost.exe 2220 Services 0 9,352 K
svchost.exe 2272 Services 0 1,352 K
svchost.exe 2304 Services 0 1,884 K
svchost.exe 2376 Services 0 1,292 K
svchost.exe 2388 Services 0 3,120 K
svchost.exe 2468 Services 0 4,148 K
svchost.exe 2736 Services 0 1,884 K
svchost.exe 2840 Services 0 2,984 K
svchost.exe 2896 Services 0 184 K
svchost.exe 2972 Services 0 1,296 K
svchost.exe 2668 Services 0 8,384 K
svchost.exe 3088 Services 0 1,480 K
svchost.exe 3096 Services 0 5,152 K
svchost.exe 3164 Services 0 1,424 K
svchost.exe 3264 Services 0 2,856 K
svchost.exe 3216 Services 0 2,836 K
Memory Compression 3244 Services 0 84,524 K
svchost.exe 3268 Services 0 9,276 K
svchost.exe 3408 Services 0 2,180 K
svchost.exe 3452 Services 0 1,380 K
svchost.exe 3584 Services 0 2,864 K
svchost.exe 3604 Services 0 3,152 K
svchost.exe 3660 Services 0 7,072 K
svchost.exe 3668 Services 0 4 K
svchost.exe 3696 Services 0 1,952 K
svchost.exe 3724 Services 0 21,432 K
svchost.exe 3816 Services 0 1,180 K
svchost.exe 3860 Services 0 7,208 K
svchost.exe 3940 Services 0 2,532 K
```

CMD

Show detailed:

Image Name	PID	Session Name	Session	Mem Usage	Status	User Name	CPU Time	Window Title
System Idle Process	0	Services	0	0 K	Unknown	NT AUTHORITY\SYSTEM	25.56/39	N/A
System	4	Services	0	132 K	Unknown	N/A	0:57:43	N/A
Secure System	92	Services	0	48,824 K	Unknown	N/A	0:00:00	N/A
lsass.exe	136	Services	0	13,352 K	Unknown	N/A	0:00:00	N/A
csrss.exe	588	Services	0	428 K	Unknown	N/A	0:00:00	N/A
csrss.exe	620	Services	0	2,168 K	Unknown	N/A	0:00:00	N/A
lsass.exe	964	Services	0	422 K	Unknown	N/A	0:00:00	N/A
csrss.exe	644	Services	0	6,408 K	Unknown	N/A	0:01:40	N/A
lsass.exe	948	Services	0	76 K	Unknown	N/A	0:00:02	N/A
csrss.exe	692	Services	0	13,728 K	Unknown	N/A	0:01:33	N/A
svchost.exe	1280	Services	0	25,652 K	Unknown	N/A	0:01:36	N/A
svchost.exe	1212	Services	0	140 K	Unknown	N/A	0:00:00	N/A
svchost.exe	1296	Services	0	0 K	Unknown	N/A	0:00:00	N/A
svchost.exe	1688	Services	0	15,420 K	Unknown	N/A	0:00:02	N/A
svchost.exe	1612	Services	0	3,560 K	Unknown	N/A	0:00:16	N/A
svchost.exe	1688	Services	0	7,172 K	Unknown	N/A	0:00:02	N/A
svchost.exe	1620	Services	0	780 K	Unknown	N/A	0:00:02	N/A
svchost.exe	1760	Services	0	7,236 K	Unknown	N/A	0:00:18	N/A
svchost.exe	1772	Services	0	2,648 K	Unknown	N/A	0:00:05	N/A
svchost.exe	1852	Services	0	2,476 K	Unknown	N/A	0:00:01	N/A
svchost.exe	1816	Services	0	7,892 K	Unknown	N/A	0:00:01	N/A
svchost.exe	1852	Services	0	2,476 K	Unknown	N/A	0:01:12	N/A
svchost.exe	1956	Services	0	8,832 K	Unknown	N/A	0:00:29	N/A
svchost.exe	1968	Services	0	5,784 K	Unknown	N/A	0:00:02	N/A
svchost.exe	1988	Services	0	4,080 K	Unknown	N/A	0:00:45	N/A
svchost.exe	1996	Services	0	7,772 K	Unknown	N/A	0:00:02	N/A
svchost.exe	788	Services	0	5,428 K	Unknown	N/A	0:00:02	N/A
svchost.exe	488	Services	0	2,436 K	Unknown	N/A	0:00:00	N/A
IntelCpMCPSvc.exe	2132	Services	0	168 K	Unknown	N/A	0:00:00	N/A
svchost.exe	2228	Services	0	15,920 K	Unknown	N/A	0:01:54	N/A
svchost.exe	2280	Services	0	1,152 K	Unknown	N/A	0:00:00	N/A
svchost.exe	2348	Services	0	980 K	Unknown	N/A	0:00:00	N/A
svchost.exe	2376	Services	0	1,152 K	Unknown	N/A	0:00:00	N/A
svchost.exe	2388	Services	0	3,172 K	Unknown	N/A	0:00:03	N/A
svchost.exe	2440	Services	0	4,760 K	Unknown	N/A	0:00:00	N/A
svchost.exe	2716	Services	0	1,368 K	Unknown	N/A	0:00:02	N/A
svchost.exe	2860	Services	0	5,480 K	Unknown	N/A	0:00:00	N/A
lsass.exe	2896	Services	0	180 K	Unknown	N/A	0:00:00	N/A
svchost.exe	2972	Services	0	1,160 K	Unknown	N/A	0:00:00	N/A
svchost.exe	2668	Services	0	18,776 K	Unknown	N/A	0:00:17	N/A
svchost.exe	1880	Services	0	1,440 K	Unknown	N/A	0:00:00	N/A
svchost.exe	1896	Services	0	7,888 K	Unknown	N/A	0:02:10	N/A
svchost.exe	1364	Services	0	1,420 K	Unknown	N/A	0:00:00	N/A
svchost.exe	1280	Services	0	2,824 K	Unknown	N/A	0:00:38	N/A
svchost.exe	2312	Services	0	2,820 K	Unknown	N/A	0:00:11	N/A
Memory Compression	2244	Services	0	1,22,388 K	Unknown	N/A	0:10:42	N/A
svchost.exe	1368	Services	0	2,156 K	Unknown	N/A	0:00:00	N/A
svchost.exe	1488	Services	0	2,168 K	Unknown	N/A	0:00:00	N/A
svchost.exe	1620	Services	0	1,180 K	Unknown	N/A	0:00:00	N/A
svchost.exe	1584	Services	0	2,860 K	Unknown	N/A	0:00:15	N/A
svchost.exe	1680	Services	0	2,380 K	Unknown	N/A	0:00:23	N/A
svchost.exe	1668	Services	0	9,840 K	Unknown	N/A	0:00:20	N/A
svchost.exe	1660	Services	0	1,940 K	Unknown	N/A	0:00:00	N/A
svchost.exe	1772	Services	0	2,140 K	Unknown	N/A	0:00:00	N/A
svchost.exe	1816	Services	0	2,108 K	Unknown	N/A	0:00:00	N/A

Identify Services

services.msc

Name	Description	Status	Startup Type	Log On As
ActiveX Installer (AInetSV)	Provides Use...	Manual	Manual	Local System
Adobe Acrobat Update Servi...	Adobe Acro...	Running	Automatic	Local System
Adobe Genuine Software Int...	Adobe Genu...	Running	Automatic	Local System
Agent Activation Runtime of...	Runtime for ...	Manual	Manual	Local System
Aggregated Data Platform S...	This service i...	Manual	Manual	Local Service
App Readiness	Gets apps re...	Manual	Manual	Local System
Application Identity	Determines ...	Manual Trigg...	Manual Trigg...	Local Service
Application Information	Facilitates th...	Running	Manual Trigg...	Local System
Application Layer Gateway S...	Provides sup...	Manual	Manual	Local Service
AppX Deployment Service (A...	Provides infr...	Running	Automatic (Tr...	Local System
Auto Time Zone Updater	Automaticall...	Manual Trigg...	Manual Trigg...	Local Service
ArCTP service	This is Audio...	Running	Manual Trigg...	Local Service
Background Intelligent Tran...	Transfers file...	Running	Automatic (De...	Local System
Background Tasks Infrastruc...	Windows inf...	Running	Automatic	Local System
Base Filtering Engine	The Base Fil...	Running	Automatic	Local Service
BitLocker Drive Encryption S...	BitLocker Dr...	Manual Trigg...	Manual Trigg...	Local System
Block Level Backup Engine S...	The WBDENGL...	Manual	Manual	Local System
Bluetooth Audio Gateway Se...	Service supp...	Running	Manual Trigg...	Local Service
Bluetooth Support Service	The Bluetoo...	Running	Manual Trigg...	Local Service
Bluetooth User Support Serv...	The Bluetoo...	Running	Manual Trigg...	Local System
Brave Elevation Service (Brav...	Provides ele...	Manual	Manual	Local System
Brave Update Service (Brave)	Keeps your ...	Automatic (De...	Automatic (De...	Local System
Brave Update Service (Brave...	Keeps your ...	Manual	Manual	Local System
Capability Access Manager S...	Provides fac...	Running	Automatic	Local System
CaptureDevice_cdbuff	Enables opti...	Manual	Manual	Local System
Cellular Time	This service ...	Manual Trigg...	Manual Trigg...	Local Service
Certificate Propagation	Copies user ...	Manual Trigg...	Manual Trigg...	Local System
Client Licence Service (ClpSvc)	Provides infr...	Running	Manual Trigg...	Local System
Clipboard User Service (cbl...	This user ser...	Running	Automatic (De...	Local System
Cloud Backup and Restore S...	Monitors th...	Manual	Manual	Local System
CNG Key Isolation	The CNG ke...	Running	Manual Trigg...	Local System
COM+ Event System	Supports Sy...	Running	Automatic	Local Service
COM+ System Application	Manages th...	Manual	Manual	Local System
Connected Devices Platform...	This service i...	Running	Automatic (De...	Local Service

CMD commands

```
C:\Users\harish> query

SERVICE_NAME: AdobeAcrobatService
DISPLAY_NAME: Adobe Acrobat Update Service
TYPE : 16  WIN32_OWN_PROCESS
STATE : 4  RUNNING
        (STOPPABLE, NOT_PAUSABLE, IGNORES_SHUTDOWN)
WIN32_EXIT_CODE : 0  (0x0)
SERVICE_EXIT_CODE : 0  (0x0)
CHECKPOINT : 0x0
WAIT_HINT : 0x0

SERVICE_NAME: AGSService
DISPLAY_NAME: Adobe Genuine Software Integrity Service
TYPE : 16  WIN32_OWN_PROCESS
STATE : 4  RUNNING
        (STOPPABLE, NOT_PAUSABLE, ACCEPTS_SHUTDOWN)
WIN32_EXIT_CODE : 0  (0x0)
SERVICE_EXIT_CODE : 0  (0x0)
CHECKPOINT : 0x0
WAIT_HINT : 0x0

SERVICE_NAME: AppInfo
DISPLAY_NAME: Application Information
TYPE : 30  WIN32
STATE : 4  RUNNING
        (STOPPABLE, NOT_PAUSABLE, IGNORES_SHUTDOWN)
WIN32_EXIT_CODE : 0  (0x0)
SERVICE_EXIT_CODE : 0  (0x0)
CHECKPOINT : 0x0
WAIT_HINT : 0x0

SERVICE_NAME: AppSvc
DISPLAY_NAME: Appx Deployment Service (AppXSVC)
TYPE : 30  WIN32
STATE : 4  RUNNING
        (STOPPABLE, NOT_PAUSABLE, ACCEPTS_SHUTDOWN)
WIN32_EXIT_CODE : 0  (0x0)
SERVICE_EXIT_CODE : 0  (0x0)
CHECKPOINT : 0x0
WAIT_HINT : 0x0

SERVICE_NAME: AudioEndpointBuilder
DISPLAY_NAME: Windows Audio Endpoint Builder
TYPE : 30  WIN32
STATE : 4  RUNNING
        (STOPPABLE, NOT_PAUSABLE, IGNORES_SHUTDOWN)
WIN32_EXIT_CODE : 0  (0x0)
SERVICE_EXIT_CODE : 0  (0x0)
CHECKPOINT : 0x0
WAIT_HINT : 0x0

SERVICE_NAME: Audiosrv
DISPLAY_NAME: Windows Audio
TYPE : 16  WIN32_OWN_PROCESS
STATE : 4  RUNNING
        (STOPPABLE, NOT_PAUSABLE, IGNORES_SHUTDOWN)
WIN32_EXIT_CODE : 0  (0x0)
SERVICE_EXIT_CODE : 0  (0x0)
CHECKPOINT : 0x0
WAIT_HINT : 0x0

SERVICE_NAME: BFE
DISPLAY_NAME: Base Filtering Engine
TYPE : 20  WIN32_SHARE_PROCESS
STATE : 4  RUNNING
        (STOPPABLE, NOT_PAUSABLE, IGNORES_SHUTDOWN)
WIN32_EXIT_CODE : 0  (0x0)
SERVICE_EXIT_CODE : 0  (0x0)
CHECKPOINT : 0x0
WAIT_HINT : 0x0
```

Check network connections:

```
C:\Users\hari6>netstat -ano

Active Connections

Proto Local Address           Foreign Address         State       PID
TCP    0.0.0.0:115             0.0.0.0:0               LISTENING   1340
TCP    0.0.0.0:405            0.0.0.0:0               LISTENING   4
TCP    0.0.0.0:2869           0.0.0.0:0               LISTENING   4
TCP    0.0.0.0:3386           0.0.0.0:0               LISTENING   5556
TCP    0.0.0.0:5080           0.0.0.0:0               LISTENING   7632
TCP    0.0.0.0:5824           0.0.0.0:0               LISTENING   5788
TCP    0.0.0.0:33860          0.0.0.0:0               LISTENING   5556
TCP    0.0.0.0:40604          0.0.0.0:0               LISTENING   602
TCP    0.0.0.0:40605          0.0.0.0:0               LISTENING   964
TCP    0.0.0.0:40606          0.0.0.0:0               LISTENING   1956
TCP    0.0.0.0:40607          0.0.0.0:0               LISTENING   2668
TCP    0.0.0.0:40608          0.0.0.0:0               LISTENING   3816
TCP    0.0.0.0:40792          0.0.0.0:0               LISTENING   664
TCP    127.0.0.1:8769         0.0.0.0:0               LISTENING   4
TCP    127.0.0.1:44804        0.0.0.0:0               LISTENING   12312
TCP    127.0.0.1:40602        127.0.0.1:40603         ESTABLISHED 5788
TCP    127.0.0.1:40603        127.0.0.1:40602         ESTABLISHED 5788
TCP    127.0.0.1:40604        127.0.0.1:40605         ESTABLISHED 5556
TCP    127.0.0.1:40605        127.0.0.1:40604         ESTABLISHED 5556
TCP    127.0.0.1:40606        127.0.0.1:40607         ESTABLISHED 5556
TCP    127.0.0.1:40607        127.0.0.1:40606         ESTABLISHED 5556
TCP    127.0.0.1:40780        127.0.0.1:40781         ESTABLISHED 5788
TCP    127.0.0.1:40781        127.0.0.1:40780         ESTABLISHED 5788
TCP    192.168.43.121:119     0.0.0.0:0               LISTENING   4
TCP    192.168.43.121:3880    76.223.81.4:803         ESTABLISHED 13880
TCP    192.168.43.121:44381   82.182.141.63:443       ESTABLISHED 12200
TCP    192.168.43.121:119     0.0.0.0:0               LISTENING   4
TCP    [*:*]:115             [*:*]:0                 LISTENING   1340
TCP    [*:*]:405             [*:*]:0                 LISTENING   4
TCP    [*:*]:2869            [*:*]:0                 LISTENING   4
TCP    [*:*]:3386            [*:*]:0                 LISTENING   5556
TCP    [*:*]:33860           [*:*]:0                 LISTENING   5788
TCP    [*:*]:33860           [*:*]:0                 LISTENING   5556
TCP    [*:*]:40604           [*:*]:0                 LISTENING   602
TCP    [*:*]:40605           [*:*]:0                 LISTENING   964
TCP    [*:*]:40606           [*:*]:0                 LISTENING   1956
TCP    [*:*]:40607           [*:*]:0                 LISTENING   2668
TCP    [*:*]:40608           [*:*]:0                 LISTENING   3816
TCP    [*:*]:40792           [*:*]:0                 LISTENING   664
TCP    [*:*]:42850           [*:*]:0                 LISTENING   6284
TCP    [*:*]:40609           [*:*]:0                 LISTENING   5260
TCP    [2401:9800::499::e5c:695e:f208:142:28b3]:5407 [2620:1ec:58:112]:443 ESTABLISHED 15596
TCP    [2401:9800::499::e5c:695e:f208:142:28b3]:4877 [2620:1ec:58:112]:443 ESTABLISHED 15596
TCP    [2401:9800::499::e5c:695e:f208:142:28b3]:13756 [2406:98c3:3180:a408:944b]:443 ESTABLISHED 14596
TCP    [2401:9800::499::e5c:695e:f208:142:28b3]:18812 [2603:1846:c86:c49:2]:443 TIME_WAIT 0
TCP    [2401:9800::499::e5c:695e:f208:142:28b3]:18813 [2603:1846:c86:c49:2]:443 TIME_WAIT 0
TCP    [2401:9800::499::e5c:695e:f208:142:28b3]:32168 [2406:98c1:3180:a401:282f]:443 ESTABLISHED 14596
TCP    [2401:9800::499::e5c:695e:f208:142:28b3]:46092 [2603:1848:318:152]:443 ESTABLISHED 14584
TCP    [2401:9800::499::e5c:695e:f208:142:28b3]:49460 [2603:1940:a86:6:11]:443 ESTABLISHED 4864
TCP    [2401:9800::499::e5c:695e:f208:142:28b3]:57925 [2602:1918:318:152]:443 ESTABLISHED 12200
UDP    0.0.0.0:500            *:*                      *:*         4176
UDP    0.0.0.0:4500           *:*                      *:*         7632
UDP    0.0.0.0:5353           *:*                      *:*         18756
UDP    0.0.0.0:5353           *:*                      *:*         15596
UDP    0.0.0.0:5353           *:*                      *:*         14596
UDP    0.0.0.0:5353           *:*                      *:*         18756
UDP    0.0.0.0:5353           *:*                      *:*         15596
```

Check startup programs:

```
C:\Users\hari6>wmic startup get caption,command
Caption                                Command
-----                                -
OneDrive                              "C:\Program Files\Microsoft OneDrive\OneDrive.exe" /background
MicrosoftEdgeAutoLaunch_CAB4CA78CFD1E2748B3D2C31460E3D99    "C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe" --win-session-start
com.todesktop.25820447d4kq915                                     C:\Users\hari6\AppData\Local\Programs\Perplexity\Perplexity.exe
Teams                                "C:\Users\hari6\AppData\Local\Microsoft\WindowsApps\MSTeams_9wekyb3d8bbwe\ms-teams.exe" asteam:system-initiated
SecurityHealth                       %windir%\system32\SecurityHealthSystray.exe
RtkAudioService                      "C:\WINDOWS\system32\DriverStore\FileRepository\realtekservice.inf_and64_3cf99e411755df38\RtkAudioService64.exe" -background

C:\Users\hari6>
```

- **LINUX WORK(Kali)**

- user accounts & privileges
- permissions (chmod/chown)
- firewall (UFW)
- processes and services
- disabling unnecessary services
- OS hardening best practices

Verify Linux system info

```
---(hari@vbox)-[~]
---$ whoami
hari

---(hari@vbox)-[~]
---$ uname -a
linux vbox 6.12.25-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.12.25-1kali1 (2025-04-30) x86_64 GNU/Linux

---(hari@vbox)-[~]
---$ cat /etc/os-release
PRETTY_NAME="Kali GNU/Linux Rolling"
NAME="Kali GNU/Linux"
VERSION_ID="2025.2"
VERSION="2025.2"
/VERSION_CODENAME=kali-rolling
ID=kali
D_LIKE=debian
DME_URL="https://www.kali.org/"
SUPPORT_URL="https://forums.kali.org/"
BUG_REPORT_URL="https://bugs.kali.org/"
NSI_COLOR="1;31"

---(hari@vbox)-[~]
---$ hostnamectl
Static hostname: vbox
Icon name: computer-vm
Chassis: vm
Machine ID: 24c0c4424654046be7380219ab5328a
Boot ID: f1a0790e24654046be7380219ab5328a
Virtualization: oracle
Operating System: Kali GNU/Linux Rolling
Kernel: Linux 6.12.25-amd64
Architecture: x86_64
Hardware Vendor: innotek GmbH
Hardware Model: VirtualBox
Firmware Version: VirtualBox
Firmware Date: Fri 2006-12-01
Firmware Age: 19y 1month 2w 6d
```

User Accounts & Groups

```
(hari@vbox)-[~]
$ cat -d: -f1 /etc/passwd | head -n 20
root
daemon
bin
sys
sync
games
man
lp
mail
news
uucp
proxy
www-data
backup
list
irc
_apt
nobody
systemd-network
dhcpcd
```

```
(hari@vbox)-[~]
$ whoami
hari
$ id
uid=1000(hari) gid=1000(hari) groups=1000(hari),4(adm),20(dialout),24(cdrom),25(floppy),27(sudo),29(audio),30(dip),44(video),46(plugdev),100(users),101(netdev),103(scanner),116(bluetooth),121(lpadm),124(wireshark),133(vboxsf),134(kabouter)
$ groups
hari adm dialout cdrom floppy sudo audio dip video plugdev users netdev scanner bluetooth lpadmin wireshark vboxsf kabouter
```

```
(hari@vbox)-[~]
$ cat /etc/group | head -n 20
root:x:0:
daemon:x:1:
bin:x:2:
sys:x:3:
adm:x:4:hari
tty:x:5:
disk:x:6:
lp:x:7:
mail:x:8:
news:x:9:
uucp:x:10:
man:x:12:
proxy:x:13:
kmem:x:15:
dialout:x:20:hari
fax:x:21:
voice:x:22:
cdrom:x:24:hari
floppy:x:25:hari
tape:x:26:
```

Understand Admin vs Standard User (sudo)

```
(hari@vbox)-[~]
$ sudo -l
[sudo] password for hari:
Matching Defaults entries for hari on vbox:
    env_reset, mail_badpass, secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin, use_pty

User hari may run the following commands on vbox:
    (ALL : ALL) ALL
```

```
(hari@vbox)-[~]
$ getent group sudo
sudo:x:27:hari
```

File Permissions

```
(hari@vbox)-[~]
$ mkdir linux-security-demo
$ cd linux-security-demo
$ echo "Cybersecurity Demo" > file1.txt
$ ls -l
total 4
-rw-rw-r-- 1 hari hari 19 Jan 20 02:47 file1.txt
$ ls
file1.txt
```

chmod practice

```
(hari@vbox)-[~/linux-security-demo]
$ chmod 600 file1.txt
$ ls -l file1.txt
-rw----- 1 hari hari 19 Jan 20 02:47 file1.txt
```

Meaning: only owner can read/write

```
(hari@vbox)-[~/linux-security-demo]
$ chmod 644 file1.txt
$ ls -l file1.txt
-rw-r--r-- 1 hari hari 19 Jan 20 02:47 file1.txt
```

Meaning: owner read/write, others only read


```

(hari@vbox)-[~/linux-security-demo]
$ chmod 700 file1.txt

(hari@vbox)-[~/linux-security-demo]
$ ls -l file1.txt
-rwx----- 1 hari hari 19 Jan 20 02:47 file1.txt

```

Meaning: only owner can access

chown practice

```

(hari@vbox)-[~/linux-security-demo]
$ sudo chown root:root file1.txt
[sudo] password for hari:

(hari@vbox)-[~/linux-security-demo]
$ ls -l file1.txt
-rwx----- 1 root root 19 Jan 20 02:47 file1.txt

```

- chmod controls permissions
- chown controls ownership
- Strong permissions reduce unauthorized access

Enable Firewall (UFW)

```

(hari@vbox)-[~/linux-security-demo]
$ ufw --version
ufw 0.36.2
Copyright 2008-2023 Canonical Ltd.

```

Check if UFW exists

```

(hari@vbox)-[~/linux-security-demo]
$ sudo ufw enable
Firewall is active and enabled on system startup

```

sudo ufw enable

```

(hari@vbox)-[~/linux-security-demo]
$ sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), disabled (routed)
New profiles: skip

```

sudo ufw status verbose

```

(hari@vbox)-[~/linux-security-demo]
$ sudo ufw allow 22/tcp
Rule added
Rule added (v6)

```

sudo ufw allow 22/tcp

Identify Running Processes

```

(hari@vbox)-[~/linux-security-demo]
$ ps aux | head -n 15
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1  0.0  0.6 24088 15196 ?        Ss   01:05   0:02 /sbin/init splash
root         2  0.0  0.0      0     0 ?        S    01:05   0:00 [kthreadd]
root         3  0.0  0.0      0     0 ?        S    01:05   0:00 [pool_workqueue_release]
root         4  0.0  0.0      0     0 ?        I<   01:05   0:00 [kworker/R-kvfree_rcu_reclaim]
root         5  0.0  0.0      0     0 ?        I<   01:05   0:00 [kworker/R-rcu_gp]
root         6  0.0  0.0      0     0 ?        I<   01:05   0:00 [kworker/R-sync_wq]
root         7  0.0  0.0      0     0 ?        I<   01:05   0:00 [kworker/R-slub_flushwq]
root         8  0.0  0.0      0     0 ?        I<   01:05   0:00 [kworker/R-netns]
root         9  0.0  0.0      0     0 ?        I    01:05   0:05 [kworker/0:0-events_freezable_pwr_efficient]
root        12  0.0  0.0      0     0 ?        I    01:05   0:00 [kworker/u8:0-ipv6_addrconf]
root        13  0.0  0.0      0     0 ?        I<   01:05   0:00 [kworker/R-mm_percpu_wq]
root        14  0.0  0.0      0     0 ?        I    01:05   0:00 [rcu_tasks_kthread]
root        15  0.0  0.0      0     0 ?        I    01:05   0:00 [rcu_tasks_rude_kthread]
root        16  0.0  0.0      0     0 ?        I    01:05   0:00 [rcu_tasks_trace_kthread]

```

```

(hari@vbox)-[~/linux-security-demo]
$ top
top - 03:38:28 up 2:33, 1 user, load average: 0.15, 0.11, 0.18
tasks: 168 total, 1 running, 167 sleeping, 0 stopped, 0 zombie
Cpu(s): 2.6 us, 2.7 sy, 0.0 ni, 94.5 id, 0.0 wa, 0.0 hi, 0.2 si, 0.0 st
Mem Mem : 2452.7 total, 1021.5 free, 775.4 used, 822.9 buff/cache
Mem Swap: 1495.0 total, 1495.0 free, 0.0 used, 1677.2 avail Mem

PID USER      PR  NI  VIRT  RES  SHR  S  %CPU  %MEM     time+ COMMAND
997 hari      20   0 215932 3168 2784 S   0.7   0.1   0:38.39 VBoxClient
1066 hari     20   0 888036 134900 92272 S   0.7   5.4   0:31.70 xfwm4
1120 hari     20   0 297172 51524 23208 S   0.7   2.1   0:31.32 wrapper-2.0
1122 hari     20   0 273968 28376 21168 S   0.7   1.1   0:23.29 wrapper-2.0
2756 hari     20   0 577716 62532 52828 S   0.7   2.5   0:15.20 qterminal
1127 hari     20   0 286444 38772 38920 S   0.3   1.5   0:06.61 wrapper-2.0
1165 root       20   0 318676 10016 8224 S   0.3   0.4   0:05.02 upowerd
4193 root       20   0      0      0 I    0.3   0.0   0:00.76 kworker/1:0-mm_percpu_wq
1 root       20   0 24888      0 S    0.0   0.0   0:02.14 systemd
2 root       20   0      0      0 S    0.0   0.0   0:00.02 kthreadd
3 root       20   0      0      0 S    0.0   0.0   0:00.00 pool_workqueue_release
4 root       20   0      0      0 I    0.0   0.0   0:00.00 kworker/R-kvfree_rcu_reclaim
5 root       20   0      0      0 I    0.0   0.0   0:00.00 kworker/R-rcu_gp
6 root       20   0      0      0 I    0.0   0.0   0:00.00 kworker/R-sync_wq
7 root       20   0      0      0 I    0.0   0.0   0:00.00 kworker/R-slub_flushwq
8 root       20   0      0      0 I    0.0   0.0   0:00.00 kworker/R-netns
12 root       20   0      0      0 I    0.0   0.0   0:00.02 kworker/u8:0-ipv6_addrconf
13 root       20   0      0      0 I    0.0   0.0   0:00.00 kworker/R-mm_percpu_wq
14 root       20   0      0      0 I    0.0   0.0   0:00.00 rcu_tasks_kthread
15 root       20   0      0      0 I    0.0   0.0   0:00.00 rcu_tasks_rude_kthread
16 root       20   0      0      0 I    0.0   0.0   0:00.00 rcu_tasks_trace_kthread
17 root       20   0      0      0 S    0.0   0.0   0:01.54 ksoftirqd/0
18 root       20   0      0      0 I    0.0   0.0   0:05.15 rcu_preempt
19 root       20   0      0      0 S    0.0   0.0   0:00.00 rcu_exp_par_gp_kthread_worker/0
20 root       20   0      0      0 S    0.0   0.0   0:00.03 rcu_exp_gp_kthread_worker
21 root       rt    0      0      0 S    0.0   0.0   0:00.19 migration/0
22 root      -51   0      0      0 S    0.0   0.0   0:00.00 idle_inject/0
23 root       20   0      0      0 S    0.0   0.0   0:00.00 cpuhp/0
24 root       20   0      0      0 S    0.0   0.0   0:00.00 cpuhp/1
25 root      -51   0      0      0 S    0.0   0.0   0:00.00 idle_inject/1
26 root       rt    0      0      0 S    0.0   0.0   0:00.71 migration/1
27 root       20   0      0      0 S    0.0   0.0   0:01.49 ksoftirqd/1
33 root       20   0      0      0 I    0.0   0.0   0:00.16 kworker/u8:0-kvfree_rcu_reclaim
34 root       20   0      0      0 S    0.0   0.0   0:00.00 kdevtmpfs
35 root       20   0      0      0 I    0.0   0.0   0:00.00 kworker/R-inet_frag_wq
36 root       20   0      0      0 S    0.0   0.0   0:00.00 kauditd
37 root       20   0      0      0 S    0.0   0.0   0:00.04 khungtaskd
38 root       20   0      0      0 S    0.0   0.0   0:00.00 oom_reaper
40 root      -20   0      0      0 I    0.0   0.0   0:00.00 kworker/R-writeback
42 root       20   0      0      0 S    0.0   0.0   0:00.71 kcompactd0
43 root       25   5      0      0 S    0.0   0.0   0:00.00 ksmd
44 root       39  19      0      0 S    0.0   0.0   0:01.57 khugepaged
45 root      -20   0      0      0 I    0.0   0.0   0:00.00 kworker/R-kintegrityd
46 root      -20   0      0      0 I    0.0   0.0   0:00.00 kworker/R-kblockd

```


Identify Running Services (systemctl)

```
(hari@vbox)-[~/linux-security-demo]
$ systemctl list-units --type=service --state=running
UNIT                                LOAD    ACTIVE SUB    DESCRIPTION
accounts-daemon.service            loaded active running Accounts Service
colord.service                     loaded active running Manage, Install and Generate Color Profiles
cron.service                       loaded active running Regular background program processing daemon
dbus.service                       loaded active running D-Bus System Message Bus
getty@tty1.service                 loaded active running Getty on tty1
haveged.service                    loaded active running Entropy Daemon based on the HAVEGE algorithm
lightdm.service                    loaded active running Light Display Manager
ModemManager.service               loaded active running Modem Manager
NetworkManager.service             loaded active running Network Manager
polkit.service                     loaded active running Authorization Manager
rtkit-daemon.service               loaded active running RealtimeKit Scheduling Policy Service
systemd-journald.service            loaded active running Journal Service
systemd-logind.service              loaded active running User Login Management
systemd-udevd.service               loaded active running Rule-based Manager for Device Events and Files
udisks2.service                    loaded active running Disk Manager
upower.service                     loaded active running Daemon for power management
user@1000.service                  loaded active running User Manager for UID 1000
virtualbox-guest-utils.service      loaded active running Virtualbox guest utils

Legend: LOAD    → Reflects whether the unit definition was properly loaded.
          ACTIVE → The high-level unit activation state, i.e. generalization of SUB.
          SUB     → The low-level unit activation state, values depend on unit type.

18 loaded units listed.
```

List running services:

```
(hari@vbox)-[~/linux-security-demo]
$ systemctl status ssh
o ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: disabled)
   Active: inactive (dead)
     Docs: man:sshd(8)
           man:sshd_config(5)
```

Check a specific service:

Check Open Ports (Network Attack Surface)

netid	State	Recv-Q	Send-Q	Local Address:Port	Peer Address:Port
-------	-------	--------	--------	--------------------	-------------------